

## Lecture 6

### Diagonalization, Cayley–Hamilton, Functions of Operators

Lecture 5 found the eigenvalues and eigenvectors of an operator and showed that, with enough independent eigenvectors, its matrix becomes diagonal. This lecture turns that observation into a working theory. We characterize exactly when an operator is *diagonalizable*, prove the *Cayley–Hamilton theorem* (every operator satisfies its own characteristic equation), and use diagonalization to compute *functions of operators*—above all the exponential  $e^{tA}$ , which solves linear differential equations and, in the form  $e^{-i\hat{H}t/\hbar}$ , generates time evolution in quantum mechanics.

The unifying idea is simple: in an eigenbasis an operator is just a list of independent scalings, so anything we can do to numbers—raise to a power, take an exponential, apply a function—we can do to the operator one eigenvalue at a time.

#### Learning objectives

By the end of this lecture you will be able to:

- ▶ Decide diagonalizability and diagonalize a matrix when possible.
- ▶ Distinguish algebraic from geometric multiplicity and use the minimal polynomial.
- ▶ Compute functions of a diagonalizable operator—powers,  $e^{tA}$ ,  $f(T)$ .
- ▶ State and apply Cayley–Hamilton and the spectral mapping rule.

#### 1 Diagonalizable operators

**Definition 6.1 (Diagonalizable).** An operator  $T \in \mathcal{L}(V)$  is *diagonalizable* if  $V$  has a basis consisting of eigenvectors of  $T$ ; equivalently, if  $\mathcal{M}(T)$  is a diagonal matrix in some basis.

The two formulations agree because the matrix of  $T$  in a basis  $v_1, \dots, v_n$  is diagonal exactly when  $Tv_k = \lambda_k v_k$  for each  $k$ , i.e. when every basis vector is an eigenvector. The following theorem collects the practical tests.

**Theorem 6.2 (Conditions for diagonalizability).** Let  $T \in \mathcal{L}(V)$  with  $\dim V = n$  and distinct eigenvalues  $\lambda_1, \dots, \lambda_m$ . The following are equivalent.

1.  $T$  is diagonalizable.
2.  $V$  has a basis of eigenvectors of  $T$ .
3.  $V = E(\lambda_1, T) \oplus \dots \oplus E(\lambda_m, T)$ .
4.  $\dim V = \dim E(\lambda_1, T) + \dots + \dim E(\lambda_m, T)$ .

*Proof.* (1) $\Leftrightarrow$ (2) is the definition. For (2) $\Rightarrow$ (4): a basis of eigenvectors splits into vectors lying in the various eigenspaces, so  $\sum_k \dim E(\lambda_k, T) \geq n$ ; the reverse inequality always holds because the eigenspaces form a direct sum (Lecture 5), giving equality. For (4) $\Rightarrow$ (3): the sum  $E(\lambda_1, T) + \dots + E(\lambda_m, T)$  is direct (Lecture 5), so its dimension is  $\sum_k \dim E(\lambda_k, T) = n = \dim V$ ; a subspace of full dimension is all of  $V$ . For (3) $\Rightarrow$ (2): concatenating bases of the eigenspaces yields a basis of  $V$  consisting of eigenvectors. ■

**Corollary 6.3 (Distinct eigenvalues suffice).** If  $T \in \mathcal{L}(V)$  has  $\dim V$  distinct eigenvalues, then  $T$  is diagonalizable.

*Proof.* The corresponding eigenvectors are independent (Lecture 5), and an independent list of length  $\dim V$  is a basis; condition (2) holds. ■

Distinctness is sufficient but not necessary: the identity has the single eigenvalue 1 yet is already diagonal. What fails for a *defective* operator is condition (4)—some eigenspace is too small.

**Example 6.4 (A diagonalizable operator).**  $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$  has distinct eigenvalues 3, 1 (Lecture 5), so by **Corollary 6.3** it is diagonalizable, with eigenvectors  $(1, 1)$ ,  $(1, -1)$  forming a basis. ◀

**Example 6.5 (A non-diagonalizable operator).** The shear  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  has only the eigenvalue 1, with  $E(1, T) = \text{span}((1, 0))$  one-dimensional. The sum of eigenspace dimensions is  $1 < 2$ , so condition (4) fails and  $T$  is not diagonalizable: there is no basis of eigenvectors. ◀

**Example 6.6 (A repeated eigenvalue that still diagonalizes).** The matrix  $A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$  has eigenvalues 3, 2, 2. The eigenspace  $E(2, A)$  solves  $(A - 2I)v = 0$ , i.e.  $v_1 + v_2 = 0$ , which is *two*-dimensional:  $E(2, A) = \text{span}((-1, 1, 0), (0, 0, 1))$ . Together with  $E(3, A) = \text{span}((1, 0, 0))$  the eigenspace dimensions sum to  $1 + 2 = 3$ , so  $A$  is diagonalizable despite the repeated eigenvalue. With  $P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ ,

$$P^{-1}AP = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \text{diag}(3, 2, 2).$$

Contrast the shear of **Example 6.5**, where the repeated eigenvalue had only a one-dimensional eigenspace. ◀

**Example 6.7 (A complete diagonalization).** Diagonalize  $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ . The characteristic polynomial  $\lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5)$  gives eigenvalues 5 and 2; solving  $(A - 5I)v = 0$  yields the eigenvector  $(1, 1)$  and  $(A - 2I)v = 0$  yields  $(1, -2)$ . Assembling these as the columns of  $P$ ,

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}, \quad P^{-1} = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \quad P^{-1}AP = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}.$$

The recipe—characteristic polynomial, eigenvectors, assemble  $P$ —works for every operator possessing a full set of eigenvectors. ◀

## 2 Diagonalization in practice

Let  $T$  on  $\mathbb{F}^n$  be diagonalizable, with eigenvectors  $v_1, \dots, v_n$  and eigenvalues  $\lambda_1, \dots, \lambda_n$ . Collect the data into

$$P = (v_1 \ \cdots \ v_n) \quad (\text{eigenvectors as columns}), \quad D = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Since  $Av_k = \lambda_k v_k$  reads  $AP = PD$  column by column, and  $P$  is invertible,

$$A = PDP^{-1}. \tag{6.1}$$

This is the change of basis of Lecture 4, specialized to an eigenbasis (**Figure 1**). Its great virtue is that powers and, soon, functions collapse to the diagonal:

$$A^k = (PDP^{-1})^k = PD^kP^{-1} = P \text{diag}(\lambda_1^k, \dots, \lambda_n^k) P^{-1},$$

the middle factors  $P^{-1}P$  telescoping away.

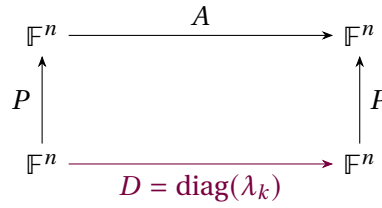


Figure 1: In the eigenbasis the operator acts as the diagonal  $D$ ; conjugating by  $P$  recovers  $A = PDP^{-1}$ . Powers and functions are applied to  $D$  entrywise.

**Example 6.8 (Computing a power).** For  $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$  with  $P = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ ,  $P^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ , and  $D = \text{diag}(3, 1)$ ,

$$A^k = P \begin{pmatrix} 3^k & 0 \\ 0 & 1 \end{pmatrix} P^{-1} = \frac{1}{2} \begin{pmatrix} 3^k + 1 & 3^k - 1 \\ 3^k - 1 & 3^k + 1 \end{pmatrix}.$$

A single diagonalization gives every power at once; computing  $A^{10}$  directly would be far more work. ◀

**Example 6.9 (Fibonacci numbers by diagonalization).** The recurrence  $F_{n+1} = F_n + F_{n-1}$  is  $\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = A \begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix}$  with  $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ , so  $\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = A^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ . The eigenvalues of  $A$  are the golden ratio  $\varphi = \frac{1+\sqrt{5}}{2}$  and  $\psi = \frac{1-\sqrt{5}}{2}$ , and  $A^n = P D^n P^{-1}$  yields Binet's closed form

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}.$$

Diagonalization turns a recurrence into a formula. ◀

**Example 6.10 (Long-run behavior of a Markov chain).** The stochastic matrix  $A = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}$  has eigenvalues 1 and 0.5. Expanding any initial state in the eigenbasis,  $A^n$  leaves the  $\lambda = 1$  component fixed and shrinks the  $\lambda = 0.5$  component by  $0.5^n \rightarrow 0$ . Hence  $A^n x_0$  converges to the  $\lambda = 1$  eigenvector—the stationary distribution (0.6, 0.4)—for any initial probability distribution  $x_0$ . Diagonalization exposes the long-run behavior immediately. ◀

**Remark 6.11 (Decoupling).** Physically, passing to the eigenbasis is *decoupling*. A system of coupled linear differential equations  $\dot{x} = Ax$  becomes, in eigen-coordinates  $y = P^{-1}x$ , the uncoupled set  $\dot{y}_k = \lambda_k y_k$ —the normal modes of a vibrating system, each oscillating independently.

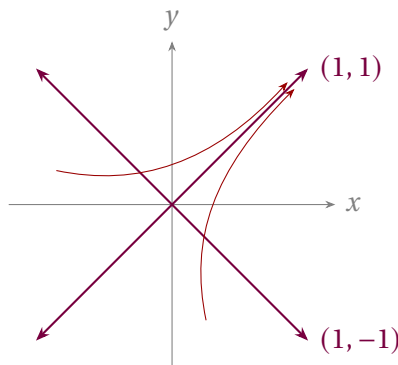


Figure 2: Phase portrait of  $\dot{x} = Ax$  for  $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$  (eigenvalues 3, 1): trajectories flow outward and bend toward the dominant eigendirection (1, 1).

**Example 6.12 (Normal modes of two coupled masses).** Two equal masses joined by identical springs obey  $\ddot{x} = -Kx$  with  $K = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$  in suitable units. The eigenvalues of  $K$  are 1 and 3, with eigenvectors  $(1, 1)$  and  $(1, -1)$ . In these *normal coordinates* the system decouples into two independent oscillators: the in-phase mode  $(1, 1)$  at angular frequency  $\omega = 1$  and the out-of-phase mode  $(1, -1)$  at  $\omega = \sqrt{3}$ . Diagonalizing  $K$  is finding the normal modes. ◀

### 3 The Cayley–Hamilton theorem

A striking fact ties an operator to its characteristic polynomial  $\chi_T(\lambda) = \det(\lambda I - T)$ .

**Theorem 6.13 (Cayley–Hamilton).** Every operator  $T$  on a finite-dimensional complex vector space satisfies its own characteristic equation:  $\chi_T(T) = 0$ .

*Proof for diagonalizable  $T$ .* Write  $A = PDP^{-1}$  with  $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ . For any polynomial  $p$ ,  $p(A) = P p(D) P^{-1}$  and  $p(D) = \text{diag}(p(\lambda_1), \dots, p(\lambda_n))$ . Taking  $p = \chi_A$  and using that each  $\lambda_k$  is a root of  $\chi_A$ ,

$$\chi_A(A) = P \text{diag}(\chi_A(\lambda_1), \dots, \chi_A(\lambda_n)) P^{-1} = P \text{diag}(0, \dots, 0) P^{-1} = 0. \quad \blacksquare$$

*Proof in general.* Over  $\mathbb{C}$  every operator has an upper-triangular matrix (Lecture 5, where triangularizability is sketched) with diagonal entries  $\lambda_1, \dots, \lambda_n$ , and then  $\chi_T(\lambda) = (\lambda - \lambda_1) \cdots (\lambda - \lambda_n)$ . The triangular-product lemma of Lecture 5 gives  $(T - \lambda_1 I) \cdots (T - \lambda_n I) = 0$  for such an operator, and that product is exactly  $\chi_T(T)$ , so  $\chi_T(T) = 0$ . ■

**Corollary 6.14 (Inverses and powers as low-degree polynomials).** If  $T$  is invertible with  $\chi_T(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0$  (so  $c_0 = (-1)^n \det T \neq 0$ ), then

$$T^{-1} = -\frac{1}{c_0}(T^{n-1} + c_{n-1}T^{n-2} + \cdots + c_1I).$$

More generally every power  $T^k$  with  $k \geq n$  reduces to a polynomial in  $T$  of degree  $< n$ .

*Proof.* Cayley–Hamilton gives  $T^n + \cdots + c_1T + c_0I = 0$ . Solve for  $c_0I$  and multiply by  $T^{-1}/(-c_0)$  for the first claim; rearranging the same relation expresses  $T^n$  (hence inductively any higher power) through lower ones. ■

**Example 6.15 (Inverse from Cayley–Hamilton).** For  $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ ,  $\chi_A(\lambda) = \lambda^2 - 4\lambda + 3$ , so  $A^2 - 4A + 3I = 0$  and  $A^{-1} = \frac{1}{3}(4I - A) = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ , which one checks satisfies  $AA^{-1} = I$ . ◀

**Example 6.16 (Cayley–Hamilton, verified by hand).** For  $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ ,  $\chi_A(\lambda) = \lambda^2 - 5\lambda - 2$ , and indeed

$$A^2 - 5A - 2I = \begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix} - \begin{pmatrix} 5 & 10 \\ 15 & 20 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

The operator annihilates its own characteristic polynomial. ◀

**Example 6.17 (Reducing a high power).** From  $A^2 = 5A + 2I$  in **Example 6.16**, every higher power collapses to a combination of  $A$  and  $I$ :  $A^3 = A \cdot A^2 = 5A^2 + 2A = 5(5A + 2I) + 2A = 27A + 10I$ , and inductively  $A^k = \alpha_k A + \beta_k I$  for scalars  $\alpha_k, \beta_k$ . Cayley–Hamilton replaces unbounded powers with arithmetic in the two “coordinates”  $I$  and  $A$ . ◀

**Remark 6.18 (Minimal polynomial).** The monic polynomial of least degree annihilating  $T$  is its *minimal polynomial*; it divides  $\chi_T$  and shares its roots (the eigenvalues). An operator is diagonalizable exactly when its minimal polynomial has no repeated roots. For the shear

$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  the minimal polynomial is  $(\lambda - 1)^2$ —since  $A - I \neq 0$  but  $(A - I)^2 = 0$ —and its repeated root flags the non-diagonalizability we found in [Example 6.5](#); whenever the eigenvalues are distinct the minimal polynomial is  $\prod_k (\lambda - \lambda_k)$  with simple roots, confirming [Corollary 6.3](#).

#### 4 Functions of operators

We have already used *polynomials* of an operator. For a diagonalizable  $T$  we can apply *any* function defined on the eigenvalues.

**Definition 6.19 (Function of a diagonalizable operator).** If  $T = PDP^{-1}$  with  $D = \text{diag}(\lambda_1, \dots, \lambda_n)$  and  $f$  is defined at each  $\lambda_k$ , set

$$f(T) = P f(D) P^{-1}, \quad f(D) = \text{diag}(f(\lambda_1), \dots, f(\lambda_n)).$$

This agrees with the power-series definition whenever the series converges, and is independent of the choice of eigenbasis.

The most important example is the *exponential*, defined for every operator by the everywhere-convergent series

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}.$$

For diagonalizable  $A = PDP^{-1}$  this sums to  $e^A = P \text{diag}(e^{\lambda_1}, \dots, e^{\lambda_n}) P^{-1}$ . Its defining property is the one that matters for dynamics.

**Proposition 6.20 (Exponential and linear flow).** For fixed  $A$ , the function  $t \mapsto e^{tA}$  satisfies  $\frac{d}{dt} e^{tA} = A e^{tA}$  and  $e^{0A} = I$ . Hence  $x(t) = e^{tA} x_0$  is the unique solution of  $\dot{x} = Ax$  with  $x(0) = x_0$ .

*Proof.* Differentiate the series term by term (it converges uniformly on bounded  $t$ -intervals):  $\frac{d}{dt} \sum_k \frac{t^k A^k}{k!} = \sum_{k \geq 1} \frac{t^{k-1} A^k}{(k-1)!} = A e^{tA}$ . Then  $\frac{d}{dt} (e^{tA} x_0) = A e^{tA} x_0$ , and  $x(0) = x_0$ . ■

**Example 6.21 (Exponential of a  $2 \times 2$  via the eigenbasis).** With  $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = PDP^{-1}$ ,  $D = \text{diag}(3, 1)$  as before,

$$e^{tA} = P \begin{pmatrix} e^{3t} & 0 \\ 0 & e^t \end{pmatrix} P^{-1} = \frac{1}{2} \begin{pmatrix} e^{3t} + e^t & e^{3t} - e^t \\ e^{3t} - e^t & e^{3t} + e^t \end{pmatrix}.$$

The solution of  $\dot{x} = Ax$  is a combination of the modes  $e^{3t}$  and  $e^t$  along the eigendirections  $(1, 1)$  and  $(1, -1)$ . ◀

**Example 6.22 (Solving an initial value problem).** For the same  $A$  with  $x(0) = (1, 0)$ , expand the initial data in the eigenbasis,  $x(0) = \frac{1}{2}(1, 1) + \frac{1}{2}(1, -1)$ , evolve each mode by its own exponential, and recombine:

$$x(t) = \frac{1}{2} e^{3t} (1, 1) + \frac{1}{2} e^t (1, -1) = \frac{1}{2} (e^{3t} + e^t, e^{3t} - e^t).$$

The dominant mode  $e^{3t} (1, 1)$  dictates the long-time behavior, and the answer agrees with the first column of  $e^{tA}$  found above. ◀

**Example 6.23 (Spin rotation:  $e^{i\theta\sigma_x}$ ).** Because  $\sigma_x^2 = I$ , the exponential series splits into even and odd powers:

$$e^{i\theta\sigma_x} = \sum_j \frac{(i\theta)^{2j}}{(2j)!} I + \sum_j \frac{(i\theta)^{2j+1}}{(2j+1)!} \sigma_x = \cos \theta I + i \sin \theta \sigma_x.$$

This is the operator that rotates a spin about the  $x$ -axis; the same calculation with any operator squaring to  $I$  produces a  $\cos/\sin$  pair. It is the two-level version of the evolution  $e^{-i\hat{H}t/\hbar}$ . ◀

**Example 6.24 (A square root of an operator).** Since  $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = PDP^{-1}$  with  $D = \text{diag}(3, 1)$  and positive eigenvalues, it has a square root

$$\sqrt{A} = P \text{diag}(\sqrt{3}, 1) P^{-1} = \frac{1}{2} \begin{pmatrix} \sqrt{3} + 1 & \sqrt{3} - 1 \\ \sqrt{3} - 1 & \sqrt{3} + 1 \end{pmatrix},$$

which satisfies  $(\sqrt{A})^2 = A$ . A function of an operator is no harder than the same function of its eigenvalues. ◀

**Remark 6.25 (Identities transport to operators).** Any scalar identity among functions holds for diagonalizable operators, since both sides act eigenvalue by eigenvalue. For instance  $\sin^2 A + \cos^2 A = I$  and  $e^A e^{-A} = I$  hold for every diagonalizable  $A$  (and, by a continuity argument, for all  $A$ ).

**Proposition 6.26 (Spectral decomposition).** If  $T$  is diagonalizable with distinct eigenvalues  $\lambda_1, \dots, \lambda_m$ , there are projections  $P_1, \dots, P_m$  (each onto an eigenspace along the others) with

$$P_i P_j = 0 \quad (i \neq j), \quad P_1 + \dots + P_m = I, \quad T = \lambda_1 P_1 + \dots + \lambda_m P_m,$$

and consequently  $f(T) = \sum_k f(\lambda_k) P_k$  for any  $f$  defined on  $\text{spec}(T)$ .

*Proof.* In the eigenbasis  $T$  is constant ( $= \lambda_k$ ) on each eigenspace block; let  $P_k$  be the projection picking out the  $E(\lambda_k, T)$  block. These satisfy the stated relations,  $T = \sum_k \lambda_k P_k$ , and applying  $f$  blockwise gives  $f(T) = \sum_k f(\lambda_k) P_k$ . ■

This is the finite-dimensional shadow of the spectral theorem of Lecture 10, in which the  $P_k$  become *orthogonal* projections.

**Example 6.27 (Spectral decomposition in coordinates).** For  $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$  the eigenprojections onto  $E(3) = \text{span}(1, 1)$  and  $E(1) = \text{span}(1, -1)$  are

$$P_1 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad P_2 = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

with  $P_1 + P_2 = I$  and  $P_1 P_2 = 0$ . Then  $A = 3P_1 + P_2$ , and **Proposition 6.26** gives  $A^k = 3^k P_1 + P_2$  and  $e^A = e^3 P_1 + e P_2$  with no matrix multiplication at all—recovering **Example 6.8** instantly. ◀

## 5 Spectral mapping and time evolution

Functions act on the spectrum in the obvious way.

**Proposition 6.28 (Spectral mapping).** If  $T$  is diagonalizable and  $f$  is defined on  $\text{spec}(T)$ , then  $\text{spec}(f(T)) = \{f(\lambda) : \lambda \in \text{spec}(T)\}$ , and every eigenvector of  $T$  for  $\lambda$  is an eigenvector of  $f(T)$  for  $f(\lambda)$ .

*Proof.* If  $Tv = \lambda v$  then  $f(T)v = Pf(D)P^{-1}v$  acts on the eigenvector  $v$  by the scalar  $f(\lambda)$ , since in eigen-coordinates  $v$  is a standard basis vector and  $f(D)$  scales it by  $f(\lambda)$ . ■

The converse inclusion for eigenvectors fails when  $f$  is not injective on the spectrum: for  $T = \text{diag}(1, -1)$  and  $f(\lambda) = \lambda^2$  we get  $f(T) = I$ , every nonzero vector of which is an eigenvector, while  $T$  has only two eigendirections. Distinct eigenvalues can be merged by  $f$ , and merging enlarges an eigenspace.

**Example 6.29 (Spectrum of a function).** For  $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$  with spectrum  $\{1, 3\}$ , the operator  $e^A$  has spectrum  $\{e, e^3\}$  and  $A^{10}$  has spectrum  $\{1, 3^{10}\}$ , all with the same eigenvectors  $(1, -1)$  and  $(1, 1)$ . Once the eigenvectors are known, no function of  $A$  requires recomputing them. ◀

**Example 6.30 (Oscillation from imaginary eigenvalues).** The generator  $A = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix}$  has eigenvalues  $\pm i\omega$ , and a short computation (or the cos/sin split, since  $A^2 = -\omega^2 I$ ) gives

$$e^{tA} = \begin{pmatrix} \cos \omega t & -\sin \omega t \\ \sin \omega t & \cos \omega t \end{pmatrix}.$$

So  $\dot{x} = Ax$  is rotation at angular frequency  $\omega$ : imaginary eigenvalues produce oscillation, exactly as real ones produced the growth and decay of [Example 6.21](#). This is the linear-algebra heart of simple harmonic motion. ◀

**Remark 6.31 (Quantum time evolution).** A quantum state evolves in time by  $|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|\psi(0)\rangle$ . In an energy eigenbasis  $\hat{H} = \text{diag}(E_1, \dots, E_n)$ , so

$$e^{-i\hat{H}t/\hbar} = \text{diag}(e^{-iE_1t/\hbar}, \dots, e^{-iE_nt/\hbar}) :$$

each stationary state merely acquires a phase  $e^{-iE_nt/\hbar}$ , and a general state is a superposition of these independently rotating phases. Diagonalizing the Hamiltonian *is* solving the dynamics—the payoff that the spectral theorem (Lecture 10) guarantees for every observable.

**Remark 6.32 (Simultaneous diagonalization).** Commuting diagonalizable operators can be brought to diagonal form in a *single* basis—the common eigenvectors of Lecture 5. Physically, compatible observables share an eigenbasis, so a system can hold simultaneously definite values for both, in sharp contrast with incompatible observables such as  $\sigma_x$  and  $\sigma_z$ , which share no eigenbasis.

## 6 When diagonalization fails

Not every operator is diagonalizable, but the generalized eigenvectors of Lecture 5 restore order. The obstruction is a *nilpotent* operator—one with  $N^k = 0$  for some  $k$ —the part of an operator that no change of basis can diagonalize away. On a single defective eigenvalue  $A = \lambda I + N$  with  $N$  nilpotent, and since  $\lambda I$  and  $N$  commute the exponential series *terminates*:

$$e^{tA} = e^{\lambda t} e^{tN} = e^{\lambda t} \left( I + tN + \frac{t^2}{2!} N^2 + \dots + \frac{t^{k-1}}{(k-1)!} N^{k-1} \right).$$

For the defective shear block  $A = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$ , the nilpotent part  $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  satisfies  $N^2 = 0$ , so

$$e^{tA} = e^{\lambda t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

The extra factor  $t$  is the signature of a defective eigenvalue: it produces solutions  $t e^{\lambda t}$ , the same ones that appear for repeated roots of a linear differential equation.

**Theorem 6.33 (Jordan normal form · cited).** Every operator on a finite-dimensional complex space has a basis of generalized eigenvectors in which its matrix is block diagonal,

each block a *Jordan block*

$$J = \lambda I + N, \quad N = \begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{pmatrix},$$

carrying the eigenvalue  $\lambda$  on the diagonal and the nilpotent shift  $N$  on the superdiagonal. Each block is one Jordan chain of generalized eigenvectors (Lecture 5).

The shift  $N$  sends each generalized eigenvector to the one below it,  $e_j \mapsto e_{j-1} \mapsto \cdots \mapsto e_1 \mapsto 0$ —a ladder that descends to zero in finitely many steps, which is exactly what nilpotency means.

**Theorem 6.34 (Jordan–Chevalley decomposition · cited).** On a complex vector space of finite dimension, every operator  $A$  splits uniquely as

$$A = D + N, \quad DN = ND,$$

with  $D$  diagonalizable and  $N$  nilpotent, and both  $D$  and  $N$  polynomials in  $A$ .

Because  $D$  and  $N$  are polynomials in  $A$ , they commute with everything  $A$  commutes with, so the splitting respects any symmetry  $A$  has—this is what makes it more than a normal-form trick. It is also the cleanest route to  $e^{tA} = e^{tD}e^{tN}$  for an arbitrary matrix: the diagonal factor supplies the oscillation and decay, the nilpotent factor the polynomial-in- $t$  corrections.

**Remark 6.35 (Nilpotents and ladder operators).** Nilpotent operators are the ladder operators of physics, seen in finite dimensions. On a spin- $j$  representation (dimension  $2j+1$ ) the raising operator  $J_+$  steps  $|m\rangle \mapsto |m+1\rangle$  until it annihilates the top state, so  $J_+^{2j+1} = 0$ :  $J_+$  is nilpotent, and its action is precisely the shift  $N$  of a Jordan block. The harmonic-oscillator lowering operator truncated to  $N$  levels is nilpotent in the same way ( $a^N = 0$ ), and the oscillator ladder of Lectures 13–14 is its infinite-dimensional limit. Nilpotency squared to zero,  $Q^2 = 0$ , is also the defining relation of the supercharge in supersymmetric quantum mechanics, of the BRST charge, and of the exterior derivative—the algebra behind cohomology.

**Remark 6.36 (Where defective operators are physical, and a note on self-study).** Self-adjointness will rule defectiveness out entirely (Lectures 9–10), so the observables of quantum mechanics are always diagonalizable. Defective operators belong instead to *non-Hermitian* physics: the critically damped oscillator, whose repeated root gives the tell-tale  $t e^{-\gamma t}$  above, and the *exceptional points* of  $PT$ -symmetric and open systems, where two eigenvectors coalesce and the Hamiltonian becomes a genuine Jordan block—a measurable effect in optics and cavity physics. For a *self-adjoint* family  $H_0 + \varepsilon V$  no such coalescence can occur at real  $\varepsilon$ ; it can at complex  $\varepsilon$ , and how near those points come to the real axis is exactly what limits the perturbation series of Lecture 10. The Jordan and Jordan–Chevalley results are offered here for self-study: we state their structure without proving the construction, and carry forward only the principle—diagonal where possible, a nilpotent correction where not.

### Summary of main concepts

An operator is diagonalizable iff it has a basis of eigenvectors, equivalently iff its eigenspace dimensions sum to  $\dim V$ ; having  $\dim V$  distinct eigenvalues is a convenient sufficient condition. In an eigenbasis  $A = PDP^{-1}$ , so powers and functions are computed one eigenvalue

at a time:  $A^k = PD^kP^{-1}$  and  $f(T) = Pf(D)P^{-1}$ , with the exponential  $e^{tA}$  solving  $\dot{x} = Ax$ . Every operator satisfies its own characteristic equation  $\chi_T(T) = 0$  (Cayley–Hamilton), which expresses inverses and high powers as low-degree polynomials in  $T$ . Spectral mapping sends eigenvalues  $\lambda$  to  $f(\lambda)$ ; in quantum mechanics this is exactly why diagonalizing the Hamiltonian solves the time evolution  $e^{-i\hat{H}t/\hbar}$ .

- ▶ **Diagonalizable**—has an eigenbasis  $\iff$  eigenspace dimensions sum to  $\dim V$ ;  $\dim V$  distinct eigenvalues suffice.
- ▶ **Diagonalized form**— $A = PDP^{-1}$ , so powers and functions act one eigenvalue at a time;  $e^{tA}$  solves  $\dot{x} = Ax$ .
- ▶ **Cayley–Hamilton**— $\chi_T(T) = 0$ ; expresses inverses and high powers as low-degree polynomials in  $T$ .
- ▶ **Spectral mapping**— $f$  sends eigenvalue  $\lambda$  to  $f(\lambda)$ —why diagonalizing  $H$  solves  $e^{-iHt/\hbar}$ .
- ▶ **Jordan form**—the canonical form when diagonalization fails (self-study).

*Exercises for this lecture are in the companion sheet `exercises-6.pdf`.*